Supplementary Material

Rehearse Before Execute: Digital Twin Action-Outcome Evaluation for UAV Recovery

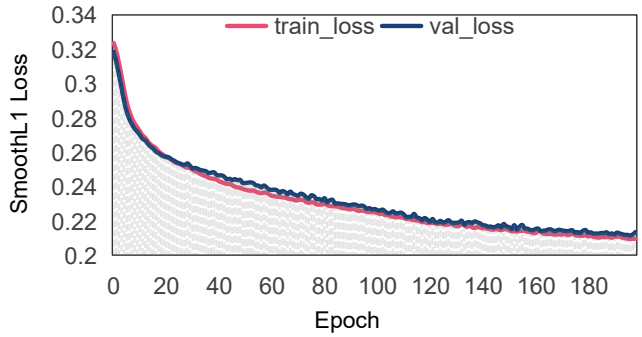


Fig. 1. **Training convergence.** A 200-epoch diagnostic run of the normal-dynamics evaluator. Action scoring uses the separately frozen 80-epoch checkpoint.

I. Learned Normal-Dynamics Evaluator

This section details the normal-dynamics evaluator: its features, architecture, training, and capacity diagnostics. The network converts candidate trajectories into evidence of deviation from nominal dynamics rather than outputting recovery actions.

A. Input features and prediction task

The aligned flight logs are resampled to 50 Hz and converted into a 20-dimensional increment-based feature vector. The features consist of first-order increments and local variation statistics, including position increments, velocity increments, speed-norm change, altitude change, acceleration increments, acceleration-norm change, angular-rate increments, angular-rate-norm change, displacement-increment norm, and short-window standard deviations of acceleration, angular rate, and speed increments. The supervised learning task is short-horizon feature prediction:

$$F_\theta : \mathbb{R}^{T\times d} \to \mathbb{R}^{K\times d}, \qquad d = 20,\ T = 50,\ K = 10.$$

Inputs and targets are robustly normalized using the median and interquartile range (IQR). Increment features emphasize local dynamic changes while reducing dependence on absolute position, mission phase, and trajectory geometry.

B. Network architecture

Table I specifies the normal-dynamics predictor. The model is a lightweight temporal convolutional predictor with two 1D convolutional layers, adaptive average pooling, and a two-layer multilayer perceptron (MLP) head. It contains 12,712 trainable parameters. The predictor estimates local feature evolution; the flight-control-compatible DT generates candidate trajectories. Keeping these roles separate allows the same compact evaluator to score all actions on a common nominal-error scale.

Algorithm 1 Normal-dynamics learning and anomaly calibration

Require: Nominal flight logs D, history window T, prediction horizon K, feature dimension d
Ensure: Predictor F_θ, robust normalizer, nominal error statistics (m_e, q_e)
1: Resample logs to a 50 Hz time grid.
2: Extract the $[0, 14.0)$ s nominal segment and construct 20-dimensional increment/statistic features.
3: Fit a median/IQR robust normalizer on the nominal features.
4: Construct prediction samples (X_t, Y_t) with stride 5.
5: Train F_θ with Adam by minimizing $\mathrm{SmoothL1}(Y_t, F_\theta(X_t))$.
6: Compute nominal prediction errors e_t and estimate $m_e = \mathrm{median}(e_t)$ and $q_e = \mathrm{IQR}(e_t)$.
7: Freeze F_θ, the normalizer, and (m_e, q_e) for candidate-consequence evaluation.

TABLE I
Architecture of the normal-dynamics predictor. Both convolutions use kernel size 3 and padding 1.

Module	Operation	Input	Output	Parameters
Input	Normalized features	50×20	50×20	0
Conv1	Conv1d + ReLU	$20 \times T$	$32 \times T$	1,952
Conv2	Conv1d + ReLU	$32 \times T$	$32 \times T$	3,104
Pooling	Adaptive average	$32 \times T$	32	0
MLP-1	Linear + ReLU	32	32	1,056
MLP-2	Linear + reshape	32	10×20	6,600
Total	Trainable parameters	–	–	**12,712**

C. Training data, loss function, and convergence

Training uses only the pre-fault nominal segments. The nominal dataset contains nine sequences, each with 700 frames, and 1,161 prediction samples. With a fixed random seed, overlapping samples are randomly split into 928 training samples and 233 validation samples. Table II reports the training configuration. The training objective is the per-sample SmoothL1 prediction error:

$$e_t = \frac{1}{Kd}\sum_{k=1}^{K}\sum_{j=1}^{d}\mathrm{SmoothL1}((Y_t)_{k,j}, F_\theta(X_t)_{k,j})\,.$$

After training, the median m_e and interquartile range q_e of nominal prediction errors are estimated from nominal samples. The model parameters θ, the normalizer, and the error statistics are then frozen for all subsequent DT rehearsals. For candidate action π, the raw anomaly uses $\epsilon = 10^{-6}$:

$$R_t^\pi = \max\left(0, \frac{e_t^\pi - m_e}{\max(q_e, \epsilon)}\right).$$

The scoring checkpoint is frozen after 80 epochs. Fig. 1 separately shows a 200-epoch diagnostic run: losses decrease rapidly and then improve more slowly.

TABLE II
TRAINING CONFIGURATION OF THE NORMAL-DYNAMICS EVALUATOR.

Item	Configuration
Nominal segment	$[0, 14.0)$ s
Target frequency	50 Hz
Past window T	50
Future horizon K	10
Stride	5
Hidden dimension	32
Checkpoint epochs	80
Diagnostic epochs	200
Batch size	64
Optimizer	Adam
Learning rate	0.001
Weight decay	0.0001

D. Capacity and window-length diagnostics

Capacity: prediction accuracy versus model size. The one-factor comparisons in Fig. 2 each use 40 training epochs. Increasing the hidden width from 16 to 32 and 64 reduces validation loss from 0.258 to 0.248 and 0.231, alongside a decrease in the 95th-percentile error. Wider networks thus fit nominal local variations more closely. However, increasing width from 32 to 64 raises the parameter count from 12,712 to 33,416, a factor of 2.63. Width 32 retains a compact model while improving on width 16, balancing accuracy and size rather than minimizing prediction loss alone.

History: longer input is not necessarily better. For history lengths of 25, 50, and 75, validation losses are 0.281, 0.248, and 0.310. Short histories contain less temporal context, whereas a longer pooled history may mix distinct local dynamics. The observed losses are consistent with this trade-off. This nonmonotonic result supports 50 samples within the tested range and motivates matching the history to local dynamics rather than simply extending its coverage.

Horizon: retaining a local evaluation scale. Prediction lengths of 5, 10, and 20 yield validation losses of 0.257, 0.248, and 0.249. The error statistics remain close for lengths 10 and 20, while the parameter count increases from 12,712 to 19,312. A 10-sample target therefore provides local-deviation evidence without the larger output head, with little error benefit from further extension in this comparison.

Remark 1 (Prediction error and score calibration). Network configuration affects both the center and tail of the error distribution. Changing capacity or windows therefore requires re-estimating nominal error statistics and freezing them before candidate comparison. Lower prediction loss alone does not imply better recovery decisions.

Algorithm 1 summarizes offline learning. Nominal data determine the feature normalization, predictor parameters, and error statistics. All three remain frozen during candidate evaluation.

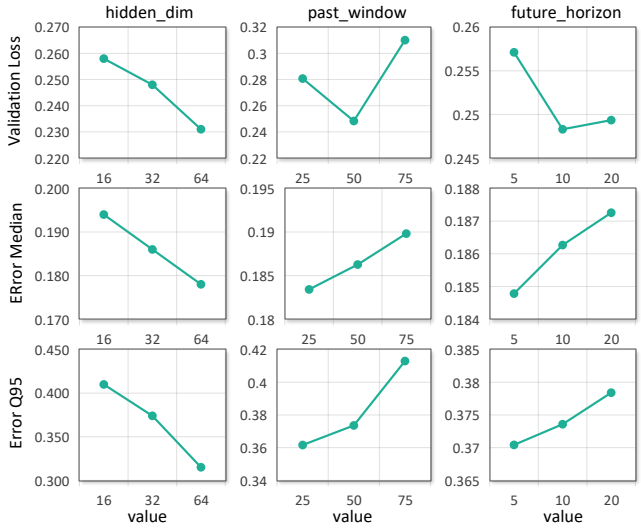


Fig. 2. **Capacity and temporal-window diagnostics.** Validation loss and error statistics under changes in hidden width, history length, and prediction horizon.

Algorithm 2 Candidate-response scoring and action selection

Require: Valid trajectories $\hat{\tau}_\pi$ for the three actions in Π, frozen evaluator, centers c_π, weight λ_s, biases b_π
Ensure: Selected action π^* and all candidate costs
for each candidate action $\pi \in \Pi$ **do**
  Extract features, apply frozen normalization, and compute R^π using F_θ and (m_e, q_e).
  Compute $J_{\text{nom}}(\pi)$ from R^π.
end for
Set $S = J_{\text{nom}}(\texttt{Continue})$.
for each candidate action $\pi \in \Pi$ **do**
  Compute $P_{\text{sev}}(\pi) = \lambda_s |S - c_\pi|$.
  Set $J(\pi) = J_{\text{nom}}(\pi) + P_{\text{sev}}(\pi) + b_\pi$.
end for
Select $\pi^* = \arg\min_{\pi \in \Pi} J(\pi)$; break exact ties in the order Continue, Return, Land.
Return π^* and all costs; dispatch only the selected command.

II. IMPLEMENTATION OF CANDIDATE-ACTION COSTS

Action selection minimizes $J(\pi)$ over the candidate set Π, comprising Continue, Return, and Land. Candidate trajectories are generated during rehearsal or loaded from archived records for retrospective evaluation. The shared, frozen evaluator converts each trajectory into a raw anomaly sequence $R^\pi = \{R^\pi_1, \ldots, R^\pi_{N_\pi}\}$. All candidates use the same feature normalization and nominal-error statistics.

A. Nominal-dynamics cost

The nominal-dynamics cost combines median anomaly, upper-tail deviation, peak deviation, and an endpoint difference:

$$J_{\text{nom}}(\pi) = 0.40\ \text{med}(R^\pi) + 0.30\, Q_{0.9}(R^\pi) + 0.10\ \max(R^\pi) - 0.20\,(R^\pi_1 - R^\pi_{N_\pi}).$$

The first three terms capture typical, upper-tail, and peak deviations from nominal dynamics. The last term rewards a reduction between the first and last valid anomaly values; it is not a fitted trajectory trend. A decreasing endpoint reduces cost.

B. Scenario-defined recovery actions

AccStd denotes the configured accelerometer-noise standard deviation. For retrospective evaluation, AccStd3, AccStd9, and AccStd15 map to Continue, Return, and Land, respectively. The operational rationale follows FAA guidance on safe operation and emergency risk reduction [1] and PX4's graded Return and Land failsafes [2].

Labels are assigned by fault-setting identifiers and shared by all trajectories and augmented records at that setting, independently of S, c_π, J, and the selected action. Consistent with the main-text tables, GT denotes the *scenario-defined recovery action*. Recovery-action agreement is the percentage of selections matching these labels. Mission completion and recovery outcomes are measured separately from execution records; they are not inferred from label agreement.

C. Flight degradation score and final cost

A low candidate-response cost can result from ending the mission early. To account for degradation while retaining the current mission, the flight degradation score is

$$S = J_{\text{nom}}(\texttt{Continue}), \qquad P_{\text{sev}}(\pi) = \lambda_s |S - c_\pi|.$$

The score S is computed from the Continue anomaly sequence. It summarizes the response under the current task and is not an initial-state value or a normalization offset. Although S is shared, the action-specific centers c_π make P_{sev} affect relative action costs.

Following the main-text notation, centers are calculated as

$$m_\pi = \text{med}_{i \in \mathcal{C}_\pi} S_i, \qquad c_\pi = \alpha e_\pi + (1-\alpha) m_\pi.$$

Here, $\mathcal{C}_\pi$ groups development records by their scenario-defined recovery action. Each group contains three original Continue records; statistical augmentations are excluded. In Continue, Return, Land order,

$$e = (140, 300, 600),$$
$$m = (133.556, 396.601, 652.898),$$
$$c = (138.711, 319.320, 610.580).$$

Remark 2 (Decision scale and execution outcomes). The centers express intervention levels on the anomaly-cost scale; they are not physical safety limits or values prescribed by operational standards. Labels and center calculation share the same intervention ordering. Execution outcomes therefore provide complementary evidence to recovery-action agreement.

Algorithm 2 specifies the scoring stage. All three valid anomaly sequences are evaluated before the final costs are compared. Offline replay uses the valid windows in each archived trajectory; online evaluation uses the collected rehearsal window. The same candidate order and exact-tie rule apply in both settings.

III. Computation and Physical Response

Table III reports a CPU benchmark using Python 3.8.1 and PyTorch 1.13.1 on Windows. Each of 300 repetitions evaluates one synthetic feature window per action, giving 900 single-window evaluations. The three-action total includes normalization, prediction-error evaluation, anomaly conversion, and cost aggregation and selection. Prediction and error calculation are already included in full evaluation. The remaining entries measure recorded command markers to subsequent home-target or mode-entry events in individual physical executions.

TABLE III
SEPARATE COMPUTATION BENCHMARK AND PHYSICAL COMMAND-RESPONSE INTERVALS. THESE MEASUREMENTS DO NOT CONSTITUTE AN ONLINE END-TO-END TIMING CHAIN.

Component / case	Measurement	Time (ms)
Prediction and error, one window	Mean / p95	0.575 / 0.724
Full evaluation, one window	Mean / p95	0.631 / 0.791
Cost aggregation / selection	Mean / p95	0.673 / 0.741
Three-action compute total	Mean / p95	**2.567 / 2.991**
A9-F8 Return	Marker to home target	363.181
A15-R Return	Marker to home target	96.179
A15-R Land	Marker to mode entry	4.398

IV. Real-flight cases and quantitative details

Table IV relates A3-C, A9-F8, and A15-R to the main-text scenarios. Table V reports the retained physical execution summaries. Duration describes the recorded interval; speed and acceleration characterize motion intensity and transients.

TABLE IV
CORRESPONDENCE BETWEEN REAL-FLIGHT CASES AND MAIN-TEXT SCENARIOS. GT DENOTES THE SCENARIO-DEFINED RECOVERY ACTION.

Case	Main scenario	GT	DT action
A3-C	VeryLight	`Continue`	`Continue`
A9-F8	TrajectoryShift	`Return`	`Return`
A15-R	Severe-Rect	`Land`	`Land`

TABLE V
PHYSICAL EXECUTION METRICS OF CANDIDATE ACTIONS IN REAL-FLIGHT CASES.

Case	Action	Duration (s)	Final altitude (m)	Max speed (m/s)	RMS speed (m/s)	Peak acc. (m/s^2)
A3-C	Continue	32.19	0.332	0.761	0.507	5.28
A3-C	Return	13.21	-0.041	1.278	0.472	26.34
A3-C	Land	10.09	0.209	0.770	0.387	3.88
A9-F8	Continue	32.09	-0.001	1.405	0.763	20.80
A9-F8	Return	13.40	-0.219	1.690	0.589	21.48
A9-F8	Land	10.29	0.041	0.907	0.406	9.10
A15-R	Continue	24.51	-1.157	1.533	0.640	54.68
A15-R	Return	13.50	0.077	2.252	0.741	46.69
A15-R	Land	10.29	-0.054	1.250	0.626	44.50

Mild degradation: task retention and intervention transients. Continue has the lowest DT cost in A3-C. Physical Return has slightly lower RMS speed than Continue but approximately five times its peak acceleration (26.34 versus 5.28 m/s^2). Lower average motion intensity can therefore coexist with stronger transients. An intervention's conservative intent does not establish smoother execution.

Moderate degradation: recovery actions have distinct responses. Return has the lowest DT cost in A9-F8. Its RMS

speed is lower than Continue's, whereas maximum speed and peak acceleration do not decrease together; Land has lower maximum speed and peak acceleration. Selection thus cannot be interpreted as minimizing every physical metric. The flight degradation score accounts for intervention need, while candidate-response costs distinguish the dynamics associated with each command.

Severe degradation: termination still entails dynamic loads. Land has the lowest DT cost in A15-R. Its physical maximum speed is lower than Return's (1.250 versus 2.252 m/s), but substantial acceleration transients remain. Ending the mission does not immediately remove degraded dynamics; the response to the recovery action also warrants evaluation.

These separately recorded executions demonstrate action-dependent responses. They complement the main-text response comparisons and retrospective rankings.

REFERENCES

[1] Federal Aviation Administration, *Small Unmanned Aircraft System (Small UAS)*, U.S. Department of Transportation, Feb. 2021, aC 107-2A, secs. 5.11 and 5.18 and app. A. Accessed: Oct. 20, 2026. [Online]. Available: https://www.faa.gov/documentLibrary/media/Advisory_Circular/AC_107-2A.pdf

[2] PX4 Development Team, *PX4 User Guide: Safety (Failsafe) Configuration*, accessed: Oct. 20, 2026. [Online]. Available: https://docs.px4.io/v1.16/en/config/safety.html